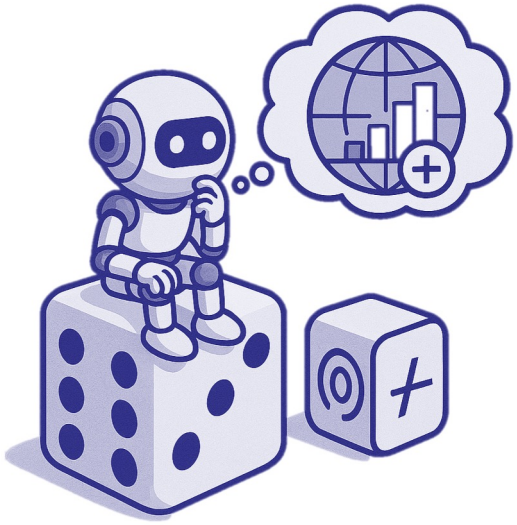
Discrete Probability for Robot Localization

From the robot on a die to the Recursive Bayes Filter





Lecture Roadmap

Six steps from uncertainty to the Bayes filter

1

Probability foundations

Experiments, events, σ-algebras, axioms, random variables

2

Reasoning with events

Conditioning, product rule, (conditional) independence

3

PMFs and belief

Distributions, joint tables, marginals, entropy

4

Inference theorems

Law of Total Probability and Bayes' rule

5

Markov models

Memory, the Markov property, chains and HMMs

6

The Bayes filter

Prediction–correction recursion and why it scales

Why Probability?

Robots act and sense under uncertainty



Motion

Commands are executed imperfectly: wheels slip and actuators err.



Sensing

Measurements are corrupted by noise and can be plainly wrong.



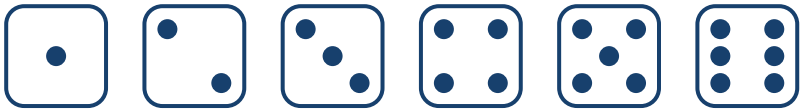
Environment

The world is only partially known, and maps are imperfect.

Consequence: localization and mapping must be formulated in a probabilistic framework. This lecture builds that framework and arrives at the Recursive Bayes Filter.

Discrete Probability

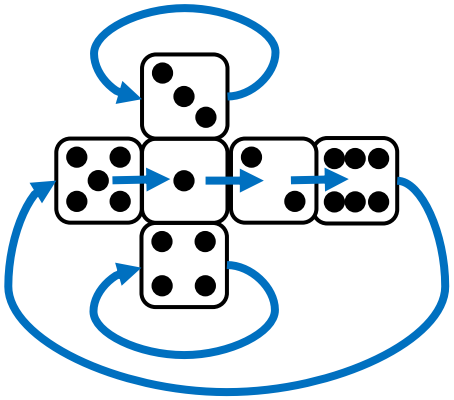
The Robot on a Die Example



Imperfect motion

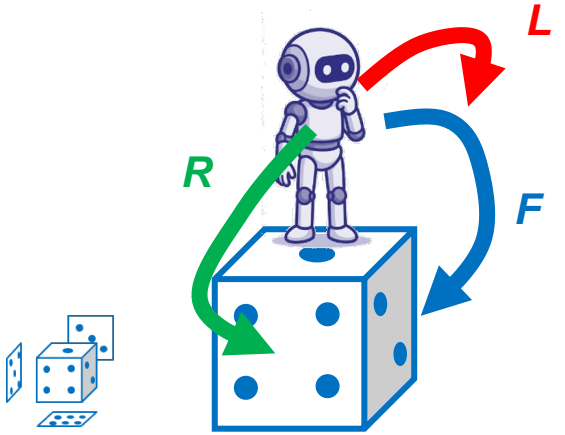
A command (forward, left, right) reaches its target face with probability 0.5; each of the other five faces gets 0.1 because of slippage.

Successful F motion



adj()

x_{k-1}	L	F	R
1	3	2	4
2	3	6	4
3	6	3	1
4	1	4	6
5	3	1	4
6	3	5	4



$$f(x_{k-1} = i, u_k = F) \Rightarrow \begin{cases} P(\bar{x}_k = adj(F, i)) = 0.5 \\ P(\bar{x}_k \notin adj(F, i) = 0.1 \end{cases}$$

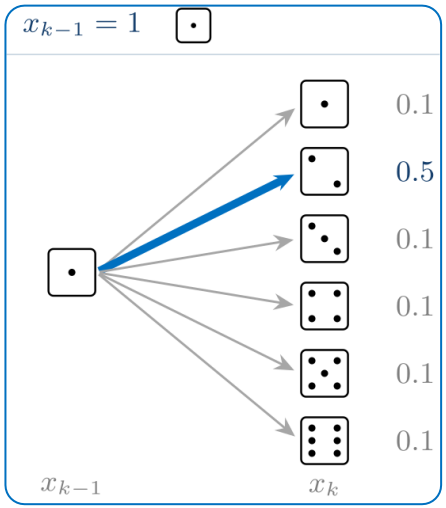
Discrete Probability

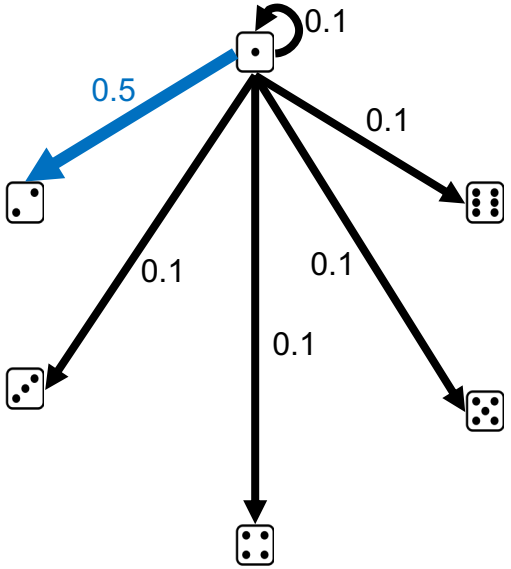
The Robot on a Die Example



1

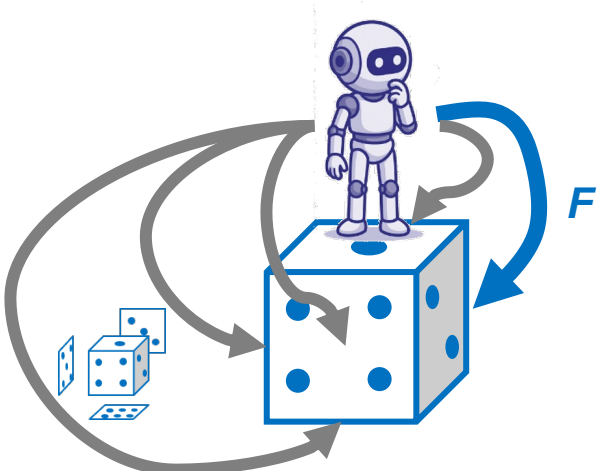
(b) Motion model from x_{k-1}.

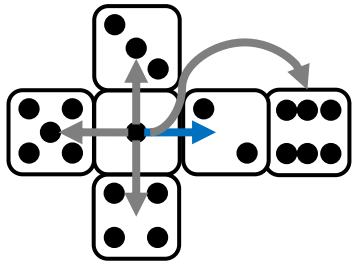




Successful F motion

Failure F motion from

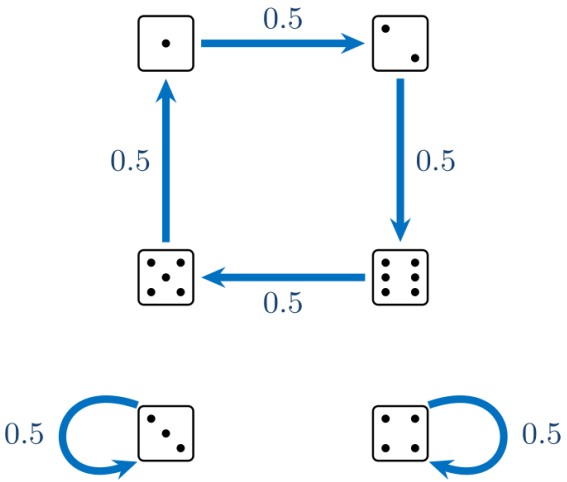




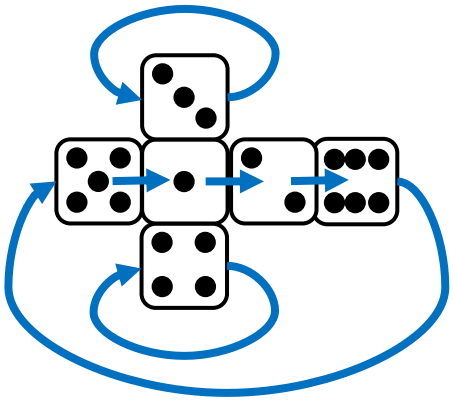
Discrete Probability

The Robot on a Die Example

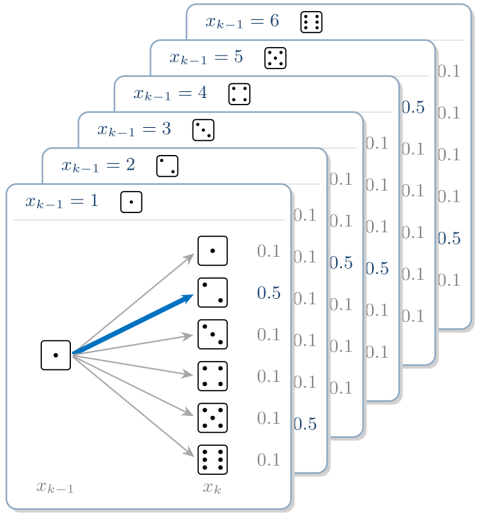
(a) Where F aims

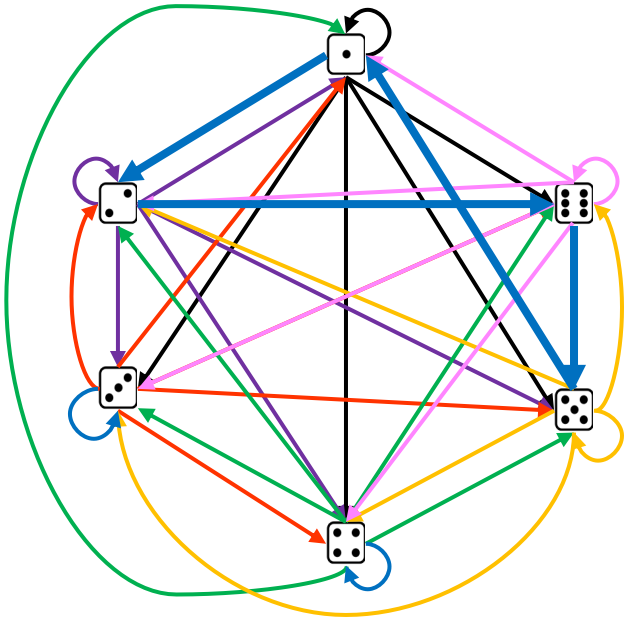


+0.1 to each of the other five faces (slippage)



(b) Motion model





Successful F motion

Failure F motion from

Failure F motion from

Failure F motion from

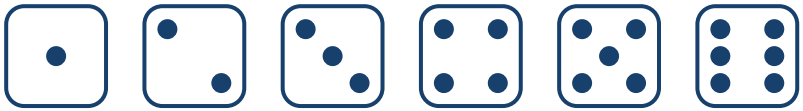
Failure F motion from

Failure F motion from

Failure F motion from

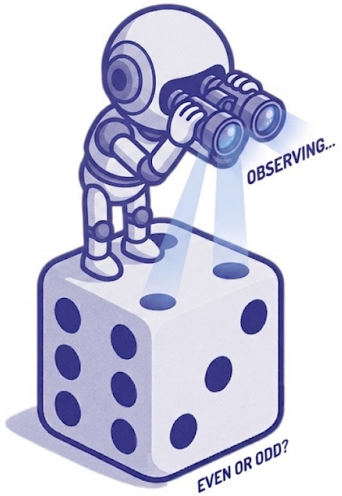
Discrete Probability

The Robot on a Die Example



Unreliable sensing

A camera reports the parity of the face (0 = even, 1 = odd): correct with probability 0.9, wrong with 0.1.



$$z_k = parity(x_k) = \begin{cases} x_k \in \{1,3,5\} \Rightarrow \begin{cases} P(z_k = 0) = 0.9 \\ P(z_k = 1) = 0.1 \end{cases} \\ x_k \in \{2,4,6\} \Rightarrow \begin{cases} P(z_k = 0) = 0.1 \\ P(z_k = 1) = 0.9 \end{cases} \end{cases}$$

Random Experiments

Three random experiments, two sample spaces

Random experiment: a procedure whose possible outcomes are known in advance, but whose actual outcome cannot be predicted with certainty.

1 Placement

Which face is the robot on?

$\Omega_X = \{\boxdot, \boxdot, \boxdot, \boxdot, \boxdot, \boxdot\}$

2 Motion

Where does it land after a command? Same sample space.

$\Omega_X = \{\boxdot, \boxdot, \boxdot, \boxdot, \boxdot, \boxdot\}$

3 Sensing

What does the camera report? (0 = even, 1 = odd)

$\Omega_Z = \{0, 1\}$

The robot is physically on exactly one face. Randomness expresses its uncertainty about which one, and later becomes its belief.

Events and Set Operations

Events are sets of outcomes, combined with set logic

An event is a subset of the sample space. It occurs when the outcome of the experiment belongs to it.

$\mathcal{A} = \{$  $\}$ even face

$\mathcal{B} = \{$  $\}$ face ≥ 5

- **Mutually exclusive:** no common outcome.
- **Partition:** mutually exclusive events that together cover Ω.

Intersection

both occur (AND)

$\mathcal{A} \cap \mathcal{B} = \{$ $\}$

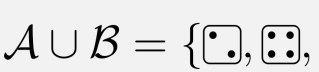
Union

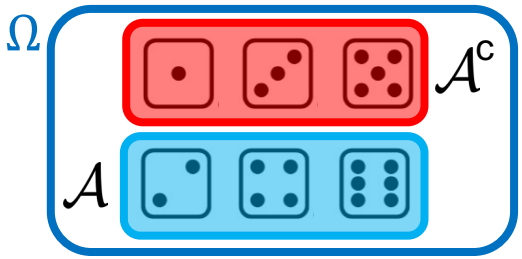
at least one occurs (OR)

$\mathcal{A} \cup \mathcal{B} = \{$ $\}$

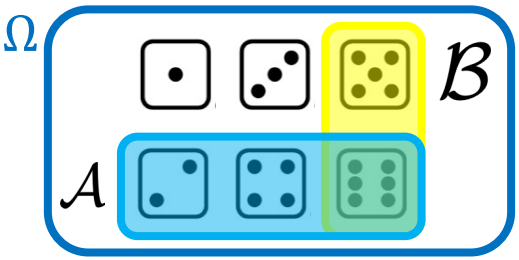
Complement

does not occur (NOT)

$\mathcal{A}^c = \{$  $\}$



Even
Odd
Face ≥ 5



σ-Algebra

The σ-algebra decides which events get a probability

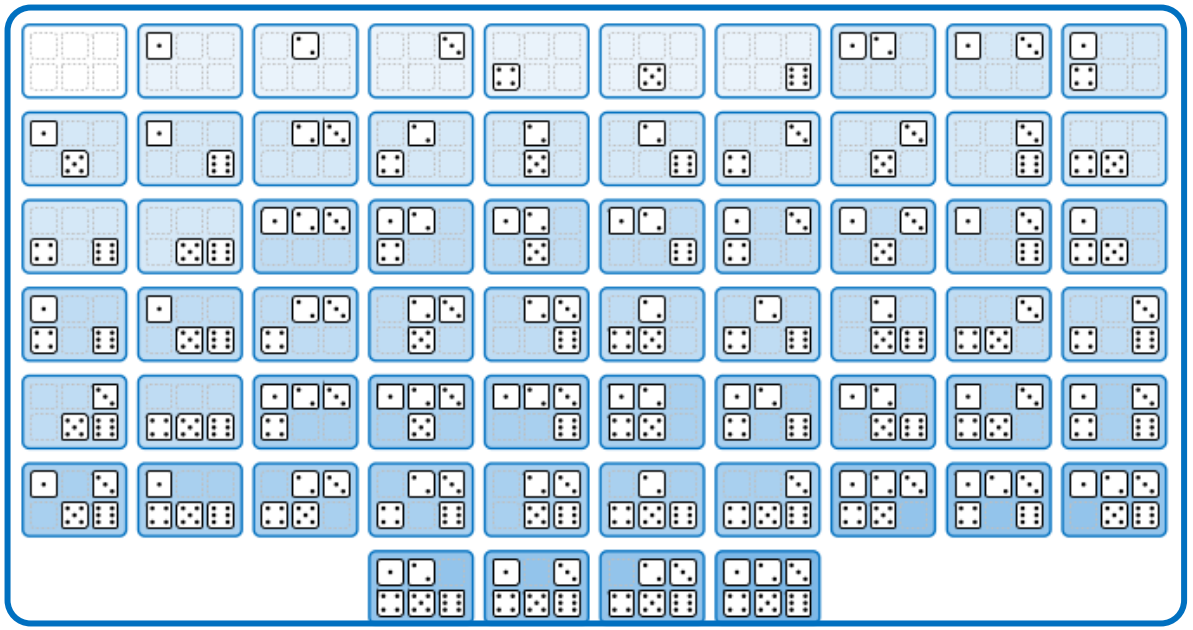
σ-algebra $\Sigma \subseteq 2^{\Omega}$: $\Omega \in \Sigma \qquad \mathcal{A} \in \Sigma \Rightarrow \mathcal{A}^c \in \Sigma \qquad \mathcal{A}_1, \mathcal{A}_2, \ldots \in \Sigma \Rightarrow \bigcup_i \mathcal{A}_i \in \Sigma$

Power set (our choice)

$\Sigma_\Omega = 2^\Omega, \qquad |2^\Omega| = 2^6 = 64$

Every group of faces gets a probability.
We use it for every finite sample space.

$\Sigma_\Omega = 2^\Omega$



σ-Algebra

The σ-algebra decides which events get a probability

σ-algebra $\Sigma \subseteq 2^{\Omega}$: $\Omega \in \Sigma$ $\quad \mathcal{A} \in \Sigma \Rightarrow \mathcal{A}^c \in \Sigma$ $\quad \mathcal{A}_1, \mathcal{A}_2, \ldots \in \Sigma \Rightarrow \bigcup_i \mathcal{A}_i \in \Sigma$

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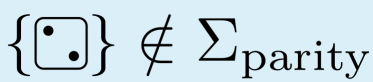
$\Sigma_{\Omega} = 2^{\Omega}, \qquad |2^{\Omega}| = 2^6 = 64$

Every group of faces gets a probability.
We use it for every finite sample space.

Parity σ-algebra (also valid)

$\Sigma_{\text{parity}} = \{\emptyset,$  $,$ $, \Omega\}$

Contains Ω, the complement of each member and every union of members.

 $\notin \Sigma_{\text{parity}}$

It cannot even express “the robot is on Face 2”. The sensor determines what the robot observes, not which events have a probability.

Probability Axioms

Kolmogorov's three axioms define a probability

1 Non-negativity

$$P(\mathcal{A}) \geq 0$$

2 Normalization

$$P(\Omega) = 1$$

3 Countable additivity

$$P\Big(\bigcup_i \mathcal{A}_i\Big) = \sum_i P(\mathcal{A}_i) \quad (\text{disjoint } \mathcal{A}_i)$$

Example: probability of an even face

$$P(\{\boxdot, \boxdot\!\boxdot, \boxdot\!\boxdot\!\boxdot\}) = P(\{\boxdot\}) + P(\{\boxdot\!\boxdot\}) + P(\{\boxdot\!\boxdot\!\boxdot\}) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$$

Probability space (Ω, Σ, P): what can happen, what can be distinguished, and how likely it is.

Properties derived from the Axioms

Everyday rules follow from the axioms

Proposition: properties of probability

1. $P(\emptyset) = 0$
2. $P(\mathcal{A}^c) = 1 - P(\mathcal{A})$
3. $\mathcal{A} \subseteq \mathcal{B} \Rightarrow P(\mathcal{A}) \leq P(\mathcal{B})$
4. $0 \leq P(\mathcal{A}) \leq 1$
5. $P(\mathcal{A} \cup \mathcal{B}) = P(\mathcal{A}) + P(\mathcal{B}) - P(\mathcal{A} \cap \mathcal{B})$

Example

$$P(\text{even} \cup \text{face} \geq 5) = \frac{3}{6} + \frac{2}{6} - \frac{1}{6} = \frac{4}{6}$$

Without the correction term, Face 6 would be counted twice.

Random variables turn outcomes into numbers

$$\boldsymbol{x}\colon \Omega \to \mathcal{X}, \qquad P(\boldsymbol{x} \in A) \triangleq P\big(\boldsymbol{x}^{-1}(A)\big)$$

The die-face random variable

$$\boldsymbol{x}\colon \Omega_X \to \{1, \ldots, 6\}$$

$$\boxed{i} \mapsto i$$

Events become pre-images

$$\boldsymbol{x}^{-1}(\{2, 4, 6\}) = \{\boxed{\cdot\,}, \boxed{\cdot\,\cdot}, \boxed{\cdot\,\cdot\,\cdot}\}$$

"The robot is on an even face."

Discrete: countable values (this chapter).

Continuous: values in a continuum such as $\mathbb{R}^n$ (next chapter).

Conditioning shrinks the sample space

Definition

$$P(\mathcal{A} \mid \mathcal{B}) = \frac{P(\mathcal{A} \cap \mathcal{B})}{P(\mathcal{B})}, \qquad P(\mathcal{B}) > 0$$

Knowing that the robot is on a face ≥ 5 …

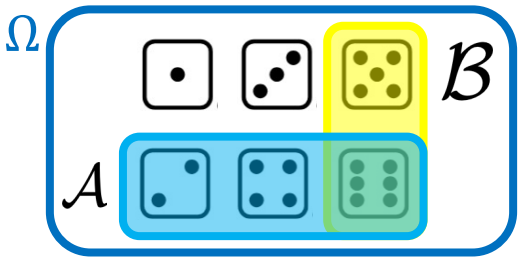


Faces 1–4 are ruled out; of the two that remain, only Face 6 is even.

$$P(\text{even} \mid \text{face} \geq 5) = \frac{P(\{\boxplus\})}{P(\{\boxdot, \boxplus\})} = \frac{1/6}{2/6} = 0.5$$

Even

Face ≥ 5



Intersection of Events: the Product Rule

The product rule turns models into joint probabilities

Corollary

$$P(\mathcal{A} \cap \mathcal{B}) = P(\mathcal{A} \mid \mathcal{B})\, P(\mathcal{B}) = P(\mathcal{B} \mid \mathcal{A})\, P(\mathcal{A})$$

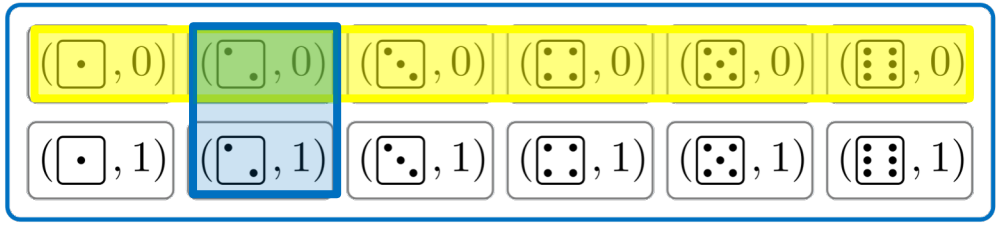
Combine both experiments into one sample space:

$$\Omega_{XZ} = \Omega_X \times \Omega_Z = \{(\boxed{i}, z)\} \quad (12 \text{ outcomes})$$

A = “robot on Face 2”, B = “sensor reports even”

$$P(\mathcal{A} \cap \mathcal{B}) = \underbrace{P(\mathcal{B} \mid \mathcal{A})}_{\text{sensor model} = 0.9}\ \underbrace{P(\mathcal{A})}_{\text{prior} = 1/6} = 0.15$$

Ω_{XZ} (12 outcomes)



Read it backwards

With P(B) = 1/2:

$$P(\mathcal{A} \mid \mathcal{B}) = \frac{P(\mathcal{A} \cap \mathcal{B})}{P(\mathcal{B})} = \frac{0.15}{0.5} = 0.30$$

Independent Events

Independence: no information flows between events

$$\mathcal{A} \perp \mathcal{B} \iff P(\mathcal{A} \cap \mathcal{B}) = P(\mathcal{A})\, P(\mathcal{B})$$
$$P(\mathcal{A}|\mathcal{B}) = P(\mathcal{A})$$

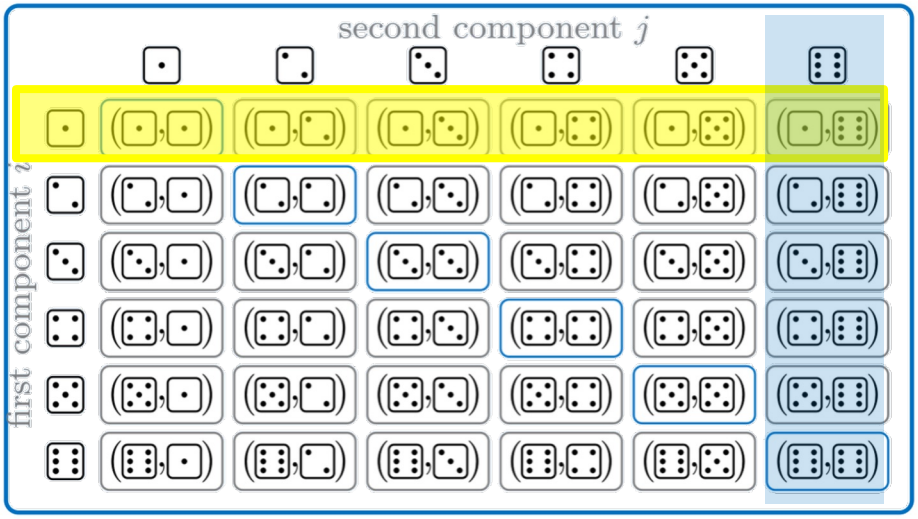
Independent: two robots, two dice



36 equally likely joint outcomes

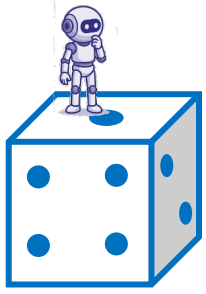
$$P(A_1 \cap B_6) = \frac{1}{36} = \frac{1}{6} \cdot \frac{1}{6}$$

One robot's face carries no information about the other's.



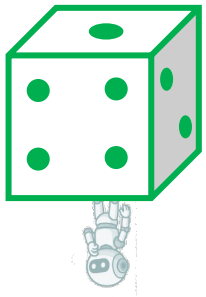
$\Omega_X \times \Omega_X$ (36 outcomes)





Robot A

Robot B



Independent Events

Independence: no information flows between events

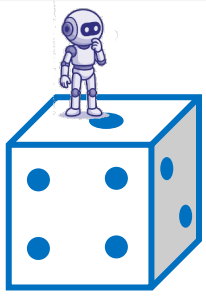
$$\mathcal{A} \perp \mathcal{B} \iff P(\mathcal{A} \cap \mathcal{B}) = P(\mathcal{A})\, P(\mathcal{B})$$
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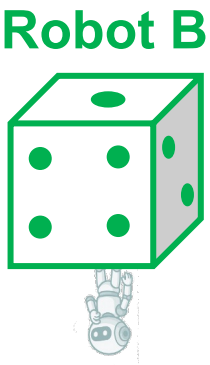
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$$P(A_1 \cap B_6) = \frac{1}{36} = \frac{1}{6} \cdot \frac{1}{6}$$

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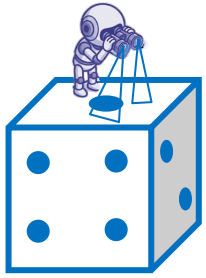


Dependent: state and parity reading

Face 1 is odd, so an odd reading becomes much more likely.

$$P(Z_1 \mid X_1) = 0.9 \neq 0.5 = P(Z_1)$$

The state causes the observation, so knowing one changes the other.



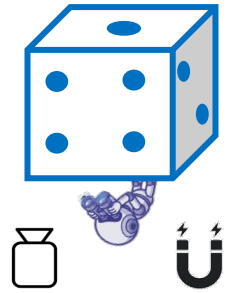
$$P(z_k = 1)\ Odd$$
$$P(x_k = \boxdot)$$

Discrete Probability

Conditional Independent Events

Two sensors are linked only through the state

$$z_C \perp z_M \mid x$$



Camera (0.9 correct) and magnetic sensor (0.8 correct), with independent errors given the state:

State unknown → dependent

$$P(C_1 \cap M_1) = (\underbrace{0.9 \cdot 0.8}_{0.72}) \cdot \tfrac{1}{2} + (\underbrace{0.1 \cdot 0.2}_{0.02}) \cdot \tfrac{1}{2} = 0.37$$

$$P(C_1)\,P(M_1) = 0.5 \cdot 0.5 = 0.25$$

An odd camera reading raises the chance of an odd magnetic reading from 0.50 to 0.74.

State known (Face 1) → independent

$$P(C_1 \cap M_1 \mid X_1) = 0.9 \cdot 0.8 = 0.72$$
$$= P(C_1 \mid X_1)\,P(M_1 \mid X_1)$$

Once the face is known, the only uncertainty left is each sensor's own error.

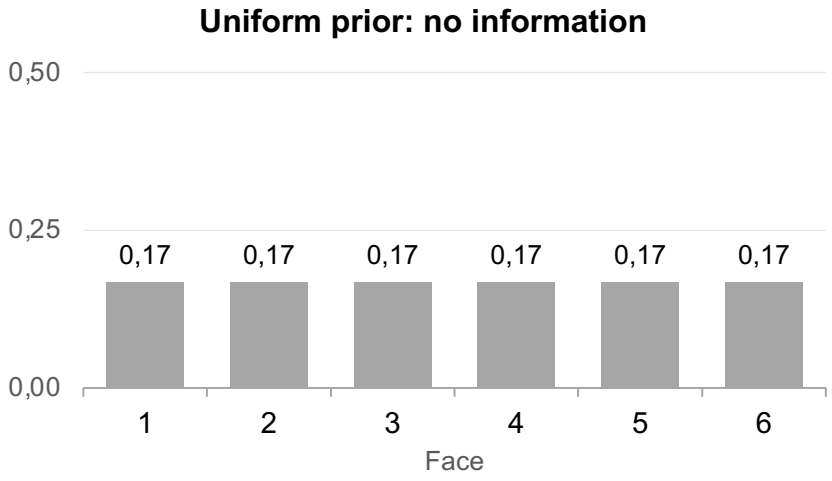
$z_C \not\perp z_M$, $\quad z_C \perp z_M \mid x$ **Correlated through a shared cause; independent once the cause is known.**

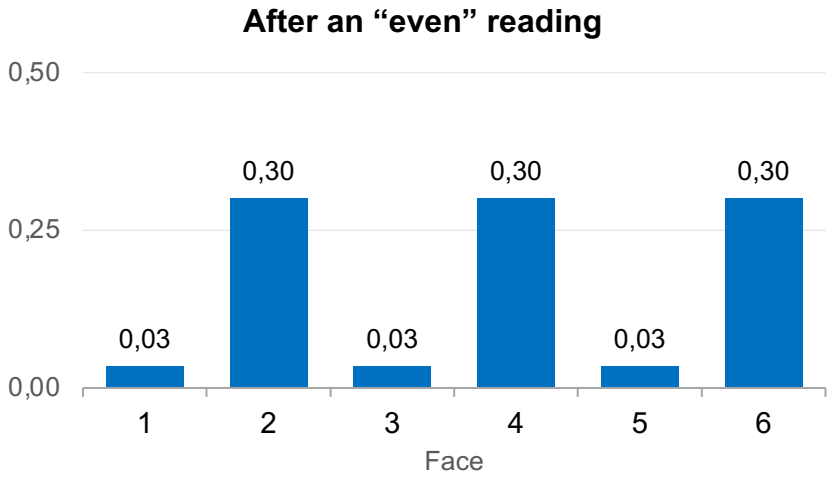
PMF and Robot Belief

The belief is a whole distribution, not a single guess

PMF $p(x) \triangleq P(\boldsymbol{x} = x), \qquad p(x) \geq 0, \qquad \sum_{x \in \mathcal{X}} p(x) = 1$

Belief $\mathrm{bel}(x) \triangleq P(\boldsymbol{x} = x \mid \mathcal{I}), \qquad \mathbf{p} \geq \mathbf{0}, \quad \mathbf{1}^\top \mathbf{p} = 1$





Joint PMF and Marginalization

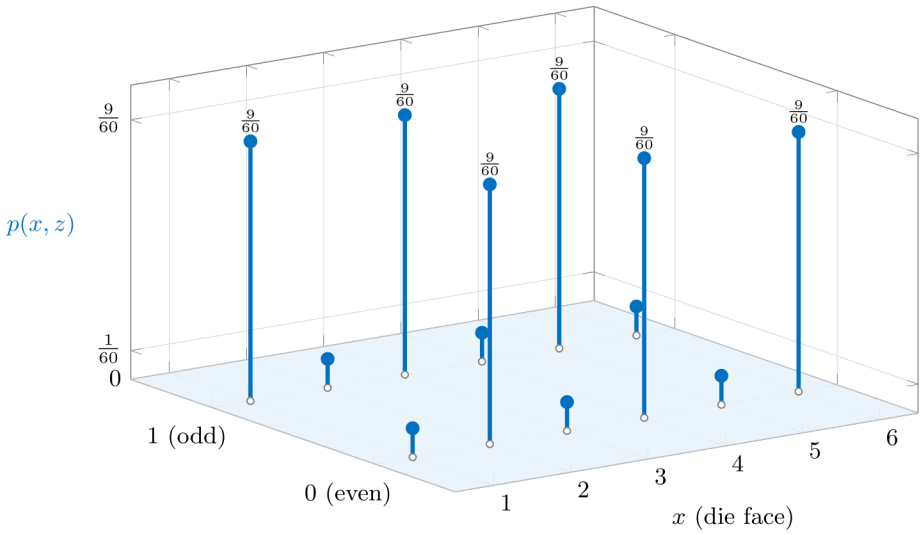
Prior × sensor model gives the joint PMF

$$p(z \mid x) = \begin{cases} 0.9, & z \text{ matches the parity of } x \\ 0.1, & \text{otherwise} \end{cases}$$

$$p(x, z) = p(z \mid x)\, p(x)$$

product rule

p(x, z)	x=1	x=2	x=3	x=4	x=5	x=6
z=0 (even)	**1/60**	9/60	**1/60**	9/60	**1/60**	9/60
z=1 (odd)	9/60	**1/60**	9/60	**1/60**	9/60	**1/60**



Joint PMF and Marginalization

Prior × sensor model gives the joint PMF

$$p(z \mid x) = \begin{cases} 0.9, & z \text{ matches the parity of } x \\ 0.1, & \text{otherwise} \end{cases}$$

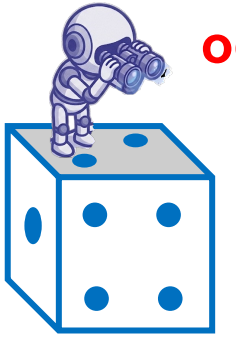
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product rule

p(x, z)	x=1	x=2	x=3	x=4	x=5	x=6
z=0 (even)	1/60	9/60	1/60	9/60	1/60	9/60
z=1 (odd)	9/60	1/60	9/60	1/60	9/60	1/60

even

odd



$$p(z = 0, x = 2) = p(z = 0 | x = 2) \cdot p(x = 2) = 0.9 \cdot \frac{1}{6}$$

$$p(z = 0, x = 1) = p(z = 0 | x = 1) \cdot p(x = 2) = 0.1 \cdot \frac{1}{6}$$

Joint PMF and Marginalization

Prior × sensor model gives the joint PMF

$$p(z \mid x) = \begin{cases} 0.9, & z \text{ matches the parity of } x \\ 0.1, & \text{otherwise} \end{cases}$$

$$p(x, z) = p(z \mid x)\, p(x)$$

product rule

p(x, z)	x=1	x=2	x=3	x=4	x=5	x=6	p(z)
z=0 (even)	1/60	9/60	1/60	9/60	1/60	9/60	1/2
z=1 (odd)	9/60	1/60	9/60	1/60	9/60	1/60	1/2
p(x)	1/6	1/6	1/6	1/6	1/6	1/6	1

Marginalization:

$$p(x) = \sum_{z} p(x, z),$$

$$p(z) = \sum_{x} p(x, z)$$

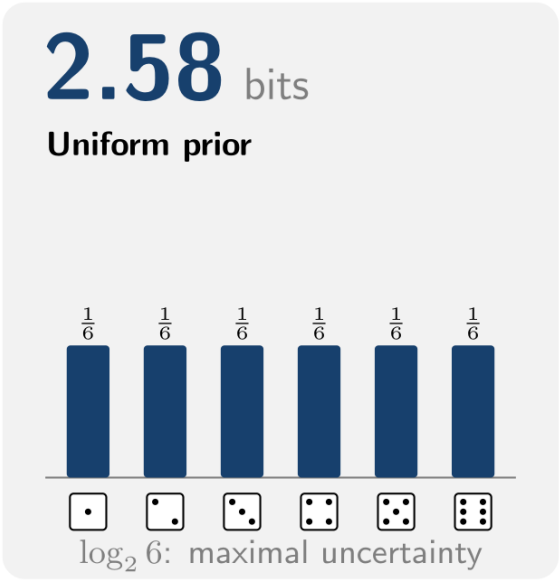
row and column sums

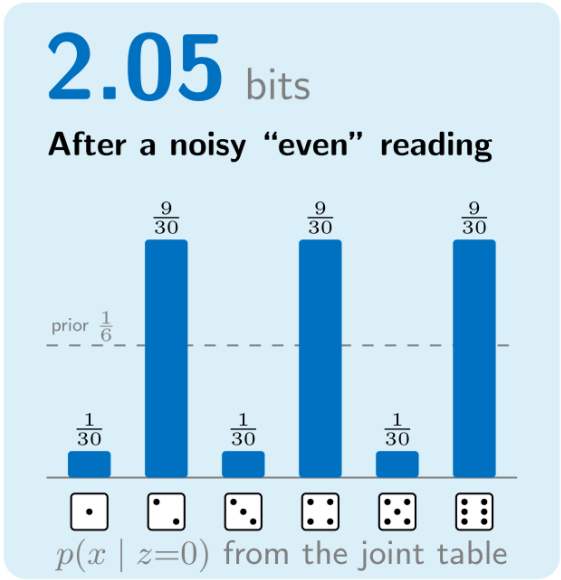
Entropy of a Discrete Belief

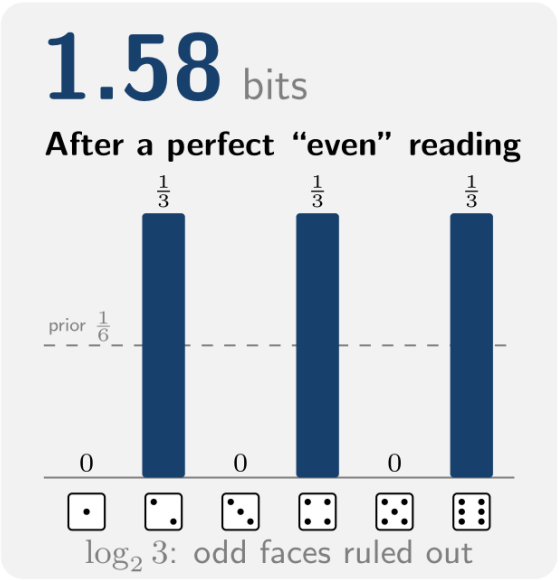
Entropy measures how uncertain the robot is

$$H(\boldsymbol{x}) = -\sum_{x} p(x) \log_2 p(x) \quad [\text{bits}], \qquad 0 \log_2 0 = 0$$

- $-\log_2 p(x)$ is the surprise of seeing x;
- entropy is the average surprise.
- It ranges from 0 (certain) to $\log_2$ n (uniform).







The Law of Total Probability sums over a partition

Theorem

$$P(\mathcal{A}) = \sum_i P(\mathcal{A} \mid \mathcal{B}_i)\, P(\mathcal{B}_i)$$

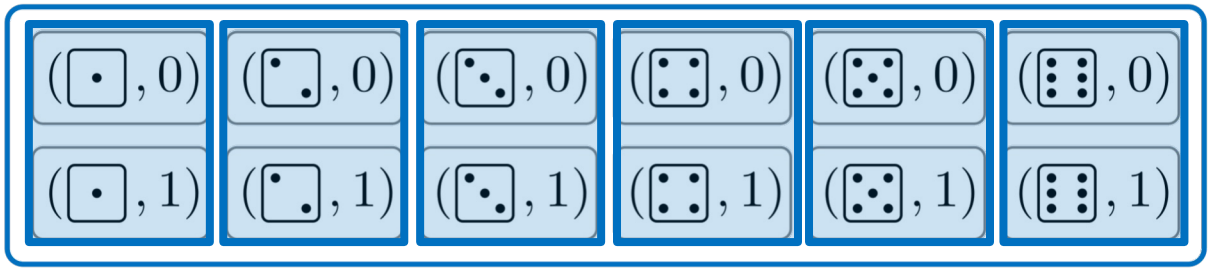
for a partition $\{B_i\}$ of Ω

$$p(z) = \sum_x p(z \mid x)\, p(x)$$

for random variables

Predicting an even reading before looking

$$p(z{=}0) =$$



Partition: x=1, x=2, x=3, x=4, x=5, x=6

The Law of Total Probability sums over a partition

Theorem

$$P(\mathcal{A}) = \sum_i P(\mathcal{A} \mid \mathcal{B}_i)\, P(\mathcal{B}_i)$$

for a partition $\{B_i\}$ of Ω

$$p(z) = \sum_x p(z \mid x)\, p(x)$$

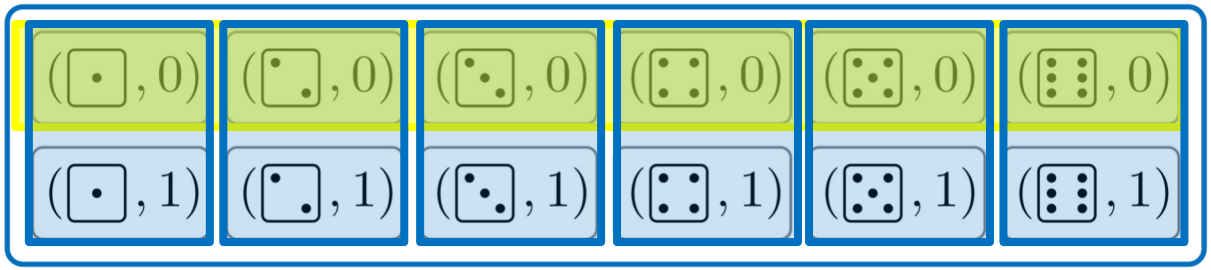
for random variables

Predicting an even reading before looking

$$p(z=0) = \sum_{i=1}^{i=6} p(z=0|x=i) \cdot p(x=i)$$

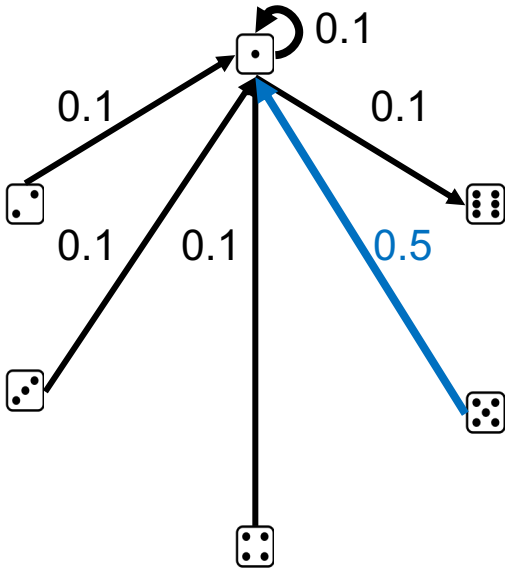
$$= 0.1 \cdot \frac{1}{6} + 0.9 \cdot \frac{1}{6} + 0.1 \cdot \frac{1}{6} + 0.9 \cdot \frac{1}{6} + 0.1 \cdot \frac{1}{6} + 0.9 \cdot \frac{1}{6} = \frac{1}{2}$$

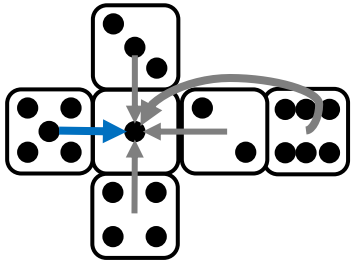
z=0



Partition: x=1, x=2, x=3, x=4, x=5, x=6

Predicting a Landing face after F Motion





$$p(x_{k-1}) = \begin{bmatrix} 0.12 & 0.12 & 0.12 & 0.12 & 0.4 & 0.12 \end{bmatrix}^\top$$

$$\begin{aligned} p(x_k = 1 | u_k = F) = & \underbrace{p(x_k = 1 | x_{k-1} = 1, u_k = F)}_{0.1} \cdot \underbrace{p(x_{k-1} = 1)}_{0.12} \\ & + \underbrace{p(x_k = 1 | x_{k-1} = 2, u_k = F)}_{0.1} \cdot \underbrace{p(x_{k-1} = 2)}_{0.12} \\ & + \underbrace{p(x_k = 1 | x_{k-1} = 3, u_k = F)}_{0.1} \cdot \underbrace{p(x_{k-1} = 3)}_{0.12} \\ & + \underbrace{p(x_k = 1 | x_{k-1} = 4, u_k = F)}_{0.1} \cdot \underbrace{p(x_{k-1} = 4)}_{0.12} \\ & + \underbrace{p(x_k = 1 | x_{k-1} = 5, u_k = F)}_{0.5} \cdot \underbrace{p(x_{k-1} = 5)}_{0.4} \\ & + \underbrace{p(x_k = 1 | x_{k-1} = 6, u_k = F)}_{0.1} \cdot \underbrace{p(x_{k-1} = 6)}_{0.12} \\ = & \ 0.26 \end{aligned}$$

Bayes' rule reverses the direction of conditioning

$$\underbrace{p(x \mid z)}_{\text{posterior}} = \frac{\overbrace{p(z \mid x)}^{\text{likelihood}} \overbrace{p(x)}^{\text{prior}}}{\underbrace{p(z)}_{\text{evidence}}} = \eta\, p(z \mid x)\, p(x) \qquad \eta = \left[\sum_{x'} p(z \mid x')\, p(x') \right]^{-1}$$

The sensor model runs from state to observation. The robot needs the opposite: given what it measured, where is it?

Bayes' rule reverses the direction of conditioning

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Correction after an even reading

1. Multiply likelihood × prior

$$p(x|z{=}0) = \eta \begin{bmatrix} 0.1 \\ 0.9 \\ 0.1 \\ 0.9 \\ 0.1 \\ 0.9 \end{bmatrix} \odot \begin{bmatrix} \frac{1}{6} \\ \frac{1}{6} \\ \frac{1}{6} \\ \frac{1}{6} \\ \frac{1}{6} \\ \frac{1}{6} \end{bmatrix} = \eta \begin{bmatrix} \frac{1}{60} \\ \frac{9}{60} \\ \frac{1}{60} \\ \frac{9}{60} \\ \frac{1}{60} \\ \frac{9}{60} \end{bmatrix}$$

2. Normalize:

$$\eta = \frac{1}{3\frac{1}{60} + 3\frac{9}{60}} = 2 \Rightarrow p(x|z{=}0) = 2 \cdot \begin{bmatrix} \frac{1}{60} \\ \frac{9}{60} \\ \frac{1}{60} \\ \frac{9}{60} \\ \frac{1}{60} \\ \frac{9}{60} \end{bmatrix} = \begin{bmatrix} \frac{0.1}{3} \\ 0.3 \\ \frac{0.1}{3} \\ 0.3 \\ \frac{0.1}{3} \\ 0.3 \end{bmatrix}$$

A process over time may depend on its whole history

Stochastic process: a collection of random variables $\{x_k\}$ on one probability space. A specific path ($1 \rightarrow 2 \rightarrow 6 \rightarrow 5$) is a realization.

In general, the next state depends on

$$p(x_{k+1} \mid x_{0:k},\ u_{1:k+1})$$

Proposition: chain rule

$$p(x_{0:k}) = p(x_0) \prod_{i=1}^{k} p(x_i \mid x_{0:i-1}, u_{0:i})$$

After an hour of motion, each factor can depend on thousands of past states. Probabilistic robotics looks for state representations under which only the present matters.

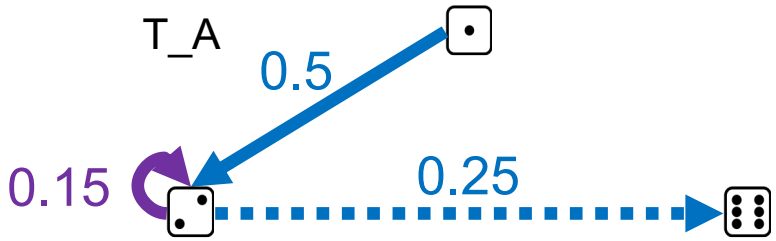
Battery decay makes the face alone a poor state

Each successful command drains the battery. With N_k the number of successes so far:

$$p(x_{k+1} = j \mid x_{0:k}, u_{k+1}) = \begin{cases} 0.5\,\gamma^{N_k}, & j = \text{target} \\ \dfrac{1 - 0.5\,\gamma^{N_k}}{5}, & \text{otherwise} \end{cases} \qquad (\gamma = 0.5)$$

Two histories under F, both on Face 2 at k = 2:

History	N_2	$p(x_3 = 6 \mid \cdot)$	Other faces
T_A = (1, 2, 2)	1	**0.25**	0.15



Non-Markovian Example: Battery Decay

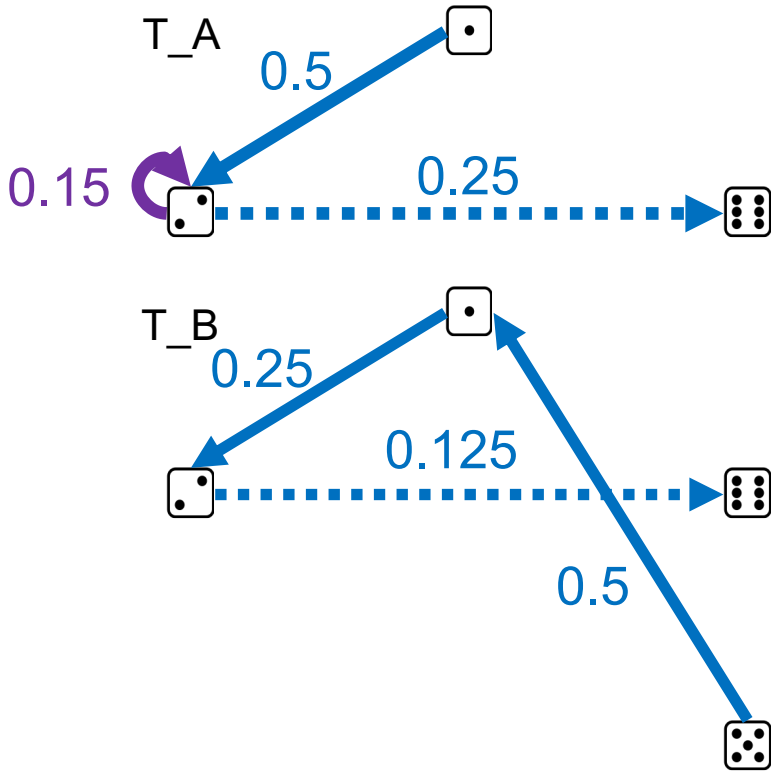
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Two histories under F, both on Face 2 at k = 2:

History	N_2	$p(x_3 = 6 \mid \cdot)$	Other faces
T_A = (1, 2, 2)	1	**0.25**	0.15
T_B = (5, 1, 2)	2	**0.125**	0.175



Same face, same command, different predictions

$$p(x_{k+1} \mid x_{0:k}, u_{k+1}) \neq p(x_{k+1} \mid x_k, u_{k+1})$$

The process is non-Markovian when the state is only the face.

Discrete Probability

Non-Markovian Example: Battery Decay

T_A = (1, 2, 2,6)

$$i=0: \quad p(x_0{=}1) = \tfrac{1}{6} \approx 0.167$$

$$i=1: \quad p(x_{0:1}{=}(1,2)) = p(x_0{=}1)\,\underbrace{p(x_1{=}2 \mid x_0{=}1, u_1{=}F)}_{\text{target, } N_0{=}0{:}\ 0.5\,\gamma^0 = 0.5} = \tfrac{1}{6}\cdot 0.5 = \tfrac{1}{12} \approx 0.083$$

$$i=2: \quad p(x_{0:2}{=}(1,2,2)) = p(x_{0:1})\,\underbrace{p(x_2{=}2 \mid x_{0:1}, u_2{=}F)}_{\text{slip, } N_1{=}1{:}\ \frac{1-0.5\,\gamma}{5} = 0.15} = \tfrac{1}{12}\cdot 0.15 = 0.0125$$

$$i=3: \quad p(x_{0:3}{=}(1,2,2,6)) = p(x_{0:2})\,\underbrace{p(x_3{=}6 \mid x_{0:2}, u_3{=}F)}_{\text{target, } N_2{=}1{:}\ 0.5\,\gamma^1 = 0.25} = 0.0125\cdot 0.25 = 0.003125$$

T_B = (5, 1, 2,6)

$$i=0: \quad p(x_0{=}5) = \tfrac{1}{6} \approx 0.167$$

$$i=1: \quad p(x_{0:1}{=}(5,1)) = p(x_0{=}5)\,\underbrace{p(x_1{=}1 \mid x_0{=}5, u_1{=}F)}_{\text{target, } N_0{=}0{:}\ 0.5\,\gamma^0 = 0.5} = \tfrac{1}{6}\cdot 0.5 = \tfrac{1}{12} \approx 0.083$$

$$i=2: \quad p(x_{0:2}{=}(5,1,2)) = p(x_{0:1})\,\underbrace{p(x_2{=}2 \mid x_{0:1}, u_2{=}F)}_{\text{target, } N_1{=}1{:}\ 0.5\,\gamma^1 = 0.25} = \tfrac{1}{12}\cdot 0.25 = \tfrac{1}{48} \approx 0.0208$$

$$i=3: \quad p(x_{0:3}{=}(5,1,2,6)) = p(x_{0:2})\,\underbrace{p(x_3{=}6 \mid x_{0:2}, u_3{=}F)}_{\text{target, } N_2{=}2{:}\ 0.5\,\gamma^2 = 0.125} = \tfrac{1}{48}\cdot 0.125 = \tfrac{1}{384} \approx 0.0026$$

The Markov property: the present summarizes the past

Definition $$p(x_k \mid x_{0:k-1},\, u_{1:k}) = p(x_k \mid x_{k-1},\, u_k)$$

Restore it by state augmentation

$$\boldsymbol{s}_k = \begin{bmatrix} f_k \\ N_k \end{bmatrix} : \qquad \begin{bmatrix} 2 \\ 1 \end{bmatrix} \neq \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

The two histories now end in different states: the memory has moved into the state. Price: a larger state space, growing with the mission.

Markovian trajectories factorize

$$p(x_{0:k} \mid u_{1:k}) = p(x_0) \prod_{i=1}^{k} p(x_i \mid x_{i-1}, u_i)$$

$$p(1,2,2,6 \mid F,F,F) = \tfrac{1}{6} \cdot 0.5 \cdot 0.1 \cdot 0.5 \approx 0.0042$$

Each factor is read directly from the motion model.

A Markov chain: the next state depends only on the current one

Definition

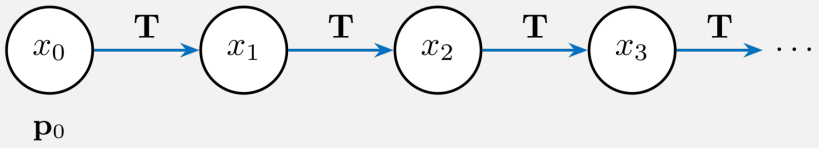
A sequence of random variables x_0, x_1, x_2, … on a state space Ω_x with the Markov property:

$$p(x_k \mid x_{0:k-1}) = p(x_k \mid x_{k-1})$$

It is fully specified by:

initial PMF $(\mathbf{p}_0)_i = p(x_0 = i)$

transition matrix $(\mathbf{T})_{ij} = p(x_k = i \mid x_{k-1} = j)$



Example: the die robot repeating F

Start on face 1 ($p_0 = e_1$); each row is p_k.

$k = 0$	1	0	0	0	0	0
$k = 1$						
$k = 2$						
$k = 3$						
$k = 4$						
$k = 5$						
$k = 6$						

?

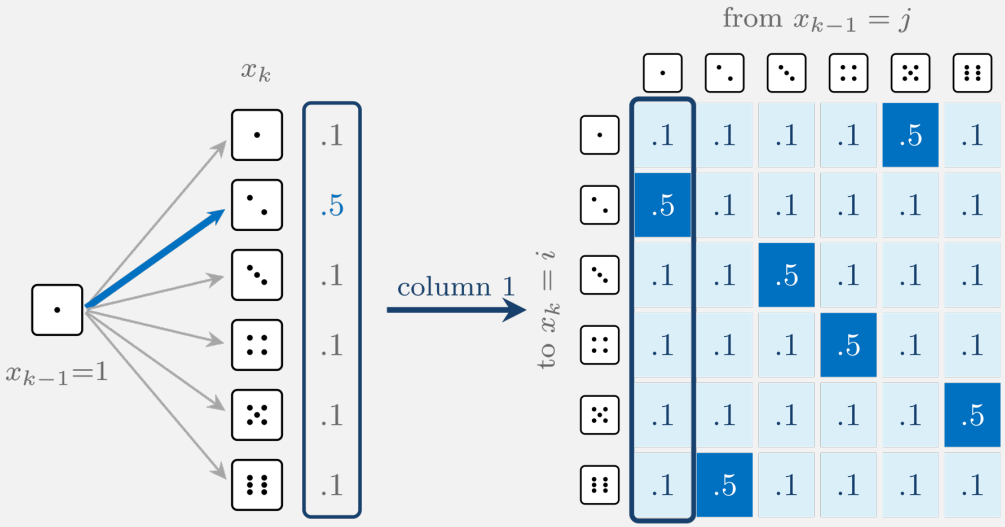
$$\mathbf{p}_k = \mathbf{T}\,\mathbf{p}_{k-1} = \mathbf{T}^k\,\mathbf{p}_0$$

Memoryless: the past matters only through the present state.

Discrete Probability

Building the Transition Matrix

Each column of the transition matrix is the motion PMF from one starting face



1 Fix a starting face j

Suppose the robot is on face j and executes F.

2 Put 0.5 on the target f(j)

The face F is aiming at: the thick blue arrow.

3 Put 0.1 on the other five

Slippage: it may land on any other face.

$$(\mathbf{T}_F)_{ij} = \begin{cases} 0.5 & i = f(j) \quad \text{(the target face)} \\ 0.1 & i \neq f(j) \quad \text{(slippage)} \end{cases}$$

Target map of F

f : ⚀↦⚂, ⚁↦⚅, ⚂↦⚂, ⚃↦⚃, ⚄↦⚀, ⚅↦⚄

$$\sum_i (\mathbf{T}_F)_{ij} = 0.5 + 5 \times 0.1 = 1 \iff \mathbf{1}^\top \mathbf{T}_F = \mathbf{1}^\top$$

Repeat for j = 1, …, 6 to fill the six columns.

A Markov chain: the next state depends only on the current one

Definition

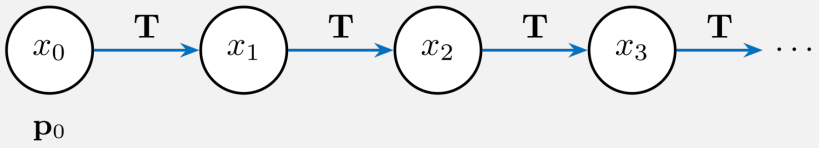
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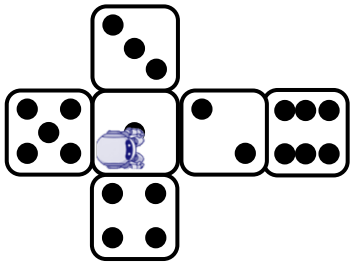
Example: the die robot repeating F

Start on face 1 ($p_0 = e_1$); each row is p_k.



$k = 0$	1	0	0	0	0	0

The peak follows 1



A Markov chain: the next state depends only on the current one

Definition

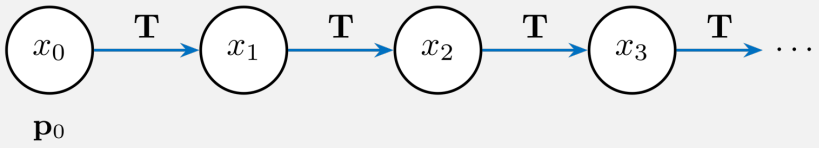
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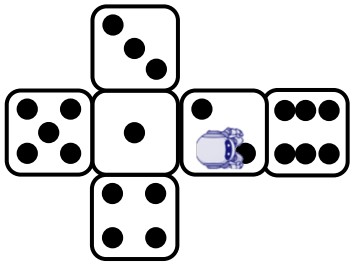
Start on face 1 ($p_0 = e_1$); each row is p_k.



$k = 0$	1	0	0	0	0	0
$k = 1$	.10	.50	.10	.10	.10	.10

The peak follows 1→2

$$\mathbf{T}_F \cdot \mathbf{p}_0 = \begin{bmatrix} 0.1 & 0.1 & 0.1 & 0.1 & 0.5 & 0.1 \\ 0.5 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.5 & 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.5 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.5 \\ 0.1 & 0.5 & 0.1 & 0.1 & 0.1 & 0.1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.1 \\ 0.5 \\ 0.1 \\ 0.1 \\ 0.1 \\ 0.1 \end{bmatrix} = \mathbf{p}_1$$



A Markov chain: the next state depends only on the current one

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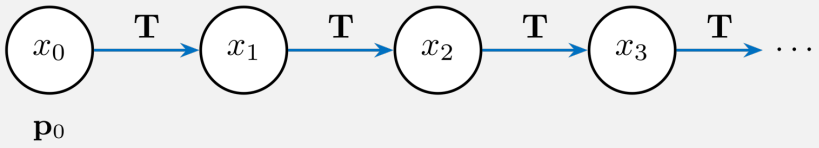
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$$p(x_k = 1|x_{k-1}) = \sum_{i=1}^{i=6} p(x_k = 1|x_{k-1} = i)p(x_{k-1} = i)$$

A Markov chain: the next state depends only on the current one

Definition

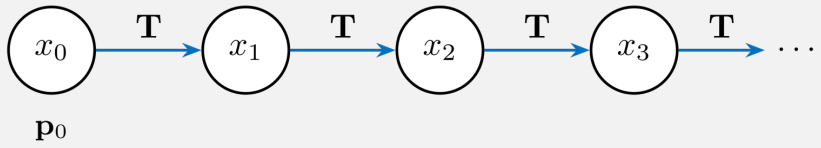
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	⚀	⚁	⚂	⚃	⚄	⚅
$k = 0$	1	0	0	0	0	0
$k = 1$	.10	.50	.10	.10	.10	.10

The peak follows 1→2

$$\mathbf{T}_F \cdot \mathbf{p}_0 = \begin{bmatrix} 0.1 & 0.1 & 0.1 & 0.1 & 0.5 & 0.1 \\ 0.5 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.5 & 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.5 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.5 \\ 0.1 & 0.5 & 0.1 & 0.1 & 0.1 & 0.1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.1 \\ 0.5 \\ 0.1 \\ 0.1 \\ 0.1 \\ 0.1 \end{bmatrix} = \mathbf{p}_1$$

$$p(x_k = 2|x_{k-1}) = \sum_{i=1}^{i=6} p(x_k = 2|x_{k-1} = i)p(x_{k-1} = i)$$

A Markov chain: the next state depends only on the current one

Definition

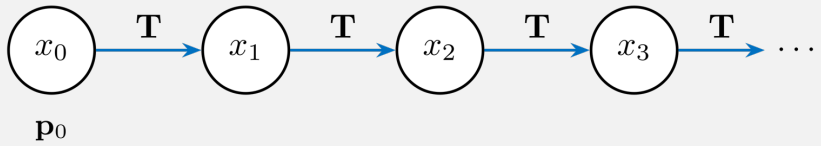
A sequence of random variables x_0, x_1, x_2, … on a state space Ω_x with the Markov property:

$$p(x_k \mid x_{0:k-1}) = p(x_k \mid x_{k-1})$$

It is fully specified by:

initial PMF $(\mathbf{p}_0)_i = p(x_0 = i)$

transition matrix $(\mathbf{T})_{ij} = p(x_k = i \mid x_{k-1} = j)$



Example: the die robot repeating F

Start on face 1 ($p_0 = e_1$); each row is p_k.



$k = 0$	1	0	0	0	0	0
$k = 1$	.10	.50	.10	.10	.10	.10

The peak follows 1→2

$$\mathbf{T}_F \cdot \mathbf{p}_0 = \begin{bmatrix} 0.1 & 0.1 & 0.1 & 0.1 & 0.5 & 0.1 \\ 0.5 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.5 & 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.5 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.5 \\ 0.1 & 0.5 & 0.1 & 0.1 & 0.1 & 0.1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.1 \\ 0.5 \\ 0.1 \\ 0.1 \\ 0.1 \\ 0.1 \end{bmatrix} = \mathbf{p}_1$$

$$\vdots$$

$$p(x_k = 6|x_{k-1}) = \sum_{i=1}^{i=6} p(x_k = 6|x_{k-1} = i)p(x_{k-1} = i)$$

A Markov chain: the next state depends only on the current one

Definition

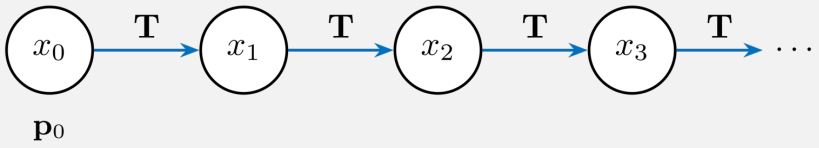
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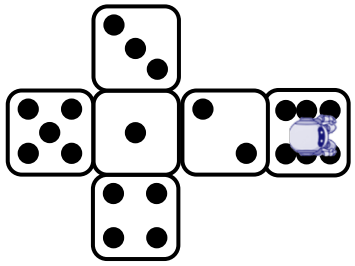
Example: the die robot repeating F

Start on face 1 ($p_0 = e_1$); each row is p_k.

$k = 0$	1	0	0	0	0	0
$k = 1$	.10	.50	.10	.10	.10	.10
$k = 2$	.14	.14	.14	.14	.14	.30

The peak follows 1→2→6

$$\mathbf{T}_F \cdot \mathbf{p}_1 = \begin{bmatrix} 0.1 & 0.1 & 0.1 & 0.1 & 0.5 & 0.1 \\ 0.5 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.5 & 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.5 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.5 \\ 0.1 & 0.5 & 0.1 & 0.1 & 0.1 & 0.1 \end{bmatrix} \cdot \begin{bmatrix} 0.1 \\ 0.5 \\ 0.1 \\ 0.1 \\ 0.1 \\ 0.1 \end{bmatrix} = \begin{bmatrix} 0.14 \\ 0.14 \\ 0.14 \\ 0.14 \\ 0.14 \\ 0.30 \end{bmatrix} = \mathbf{p}_2$$



Discrete Probability

Markov Chains

A Markov chain: the next state depends only on the current one

Definition

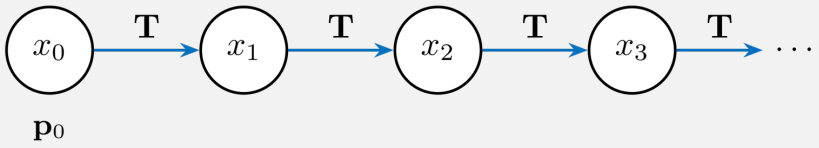
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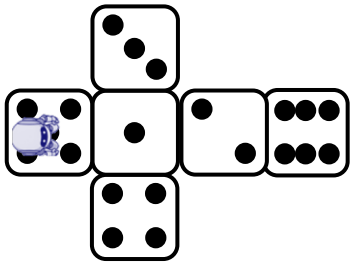
Example: the die robot repeating F

Start on face 1 ($p_0 = e_1$); each row is p_k.

						
$k = 0$	1	0	0	0	0	0
$k = 1$	.10	.50	.10	.10	.10	.10
$k = 2$	.14	.14	.14	.14	.14	.30
$k = 3$	.16	.16	.16	.16	.22	.16

The peak follows 1→2→6→5

$$\mathbf{T}_F \cdot \mathbf{p}_2 = \begin{bmatrix} 0.1 & 0.1 & 0.1 & 0.1 & 0.5 & 0.1 \\ 0.5 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.5 & 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.5 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.5 \\ 0.1 & 0.5 & 0.1 & 0.1 & 0.1 & 0.1 \end{bmatrix} \cdot \begin{bmatrix} 0.14 \\ 0.14 \\ 0.14 \\ 0.14 \\ 0.14 \\ 0.30 \end{bmatrix} = \begin{bmatrix} 0.16 \\ 0.16 \\ 0.16 \\ 0.16 \\ 0.22 \\ 0.16 \end{bmatrix} = \mathbf{p}_3$$



Markov Chains

A Markov chain: the next state depends only on the current one

Definition

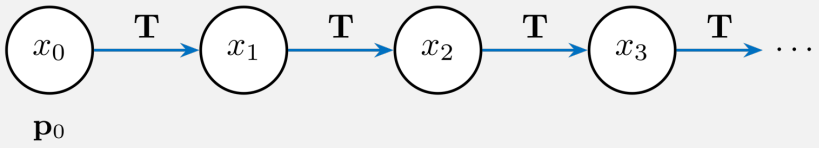
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$$p(x_k \mid x_{0:k-1}) = p(x_k \mid x_{k-1})$$

It is fully specified by:

initial PMF $(\mathbf{p}_0)_i = p(x_0 = i)$

transition matrix $(\mathbf{T})_{ij} = p(x_k = i \mid x_{k-1} = j)$



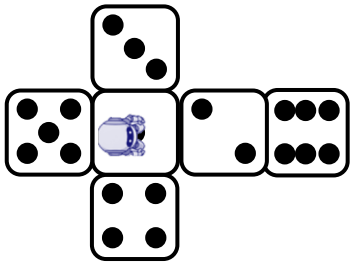
Example: the die robot repeating F

Start on face 1 ($p_0 = e_1$); each row is p_k.

						
$k = 0$	1	0	0	0	0	0
$k = 1$	.10	.50	.10	.10	.10	.10
$k = 2$	.14	.14	.14	.14	.14	.30
$k = 3$	.16	.16	.16	.16	.22	.16
$k = 4$	.19	.16	.16	.16	.16	.16

The peak follows 1→2→6→5→1

$$\mathbf{T}_F \cdot \mathbf{p}_3 = \begin{bmatrix} 0.1 & 0.1 & 0.1 & 0.1 & 0.5 & 0.1 \\ 0.5 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.5 & 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.5 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.5 \\ 0.1 & 0.5 & 0.1 & 0.1 & 0.1 & 0.1 \end{bmatrix} \cdot \begin{bmatrix} 0.16 \\ 0.16 \\ 0.16 \\ 0.16 \\ 0.22 \\ 0.16 \end{bmatrix} = \begin{bmatrix} 0.19 \\ 0.16 \\ 0.16 \\ 0.16 \\ 0.16 \\ 0.16 \end{bmatrix} = \mathbf{p}_4$$



A Markov chain: the next state depends only on the current one

Definition

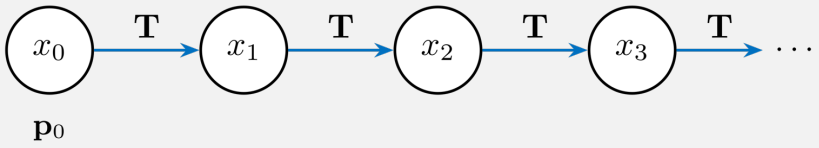
A sequence of random variables x_0, x_1, x_2, … on a state space Ω_x with the Markov property:

$$p(x_k \mid x_{0:k-1}) = p(x_k \mid x_{k-1})$$

It is fully specified by:

initial PMF $(\mathbf{p}_0)_i = p(x_0 = i)$

transition matrix $(\mathbf{T})_{ij} = p(x_k = i \mid x_{k-1} = j)$



Example: the die robot repeating F

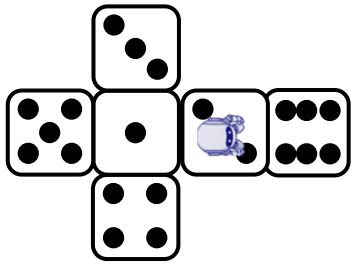
Start on face 1 ($p_0 = e_1$); each row is p_k.



$k = 0$	1	0	0	0	0	0
$k = 1$	.10	.50	.10	.10	.10	.10
$k = 2$	.14	.14	.14	.14	.14	.30
$k = 3$	.16	.16	.16	.16	.22	.16
$k = 4$	.19	.16	.16	.16	.16	.16
$k = 5$	.16	.18	.16	.16	.16	.16

The peak follows 1→2→6→5→1 →2.

$$\mathbf{T}_F \cdot \mathbf{p}_4 = \begin{bmatrix} 0.1 & 0.1 & 0.1 & 0.1 & 0.5 & 0.1 \\ 0.5 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.5 & 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.5 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.5 \\ 0.1 & 0.5 & 0.1 & 0.1 & 0.1 & 0.1 \end{bmatrix} \cdot \begin{bmatrix} 0.19 \\ 0.16 \\ 0.16 \\ 0.16 \\ 0.16 \\ 0.16 \end{bmatrix} = \begin{bmatrix} 0.16 \\ 0.18 \\ 0.16 \\ 0.16 \\ 0.16 \\ 0.16 \end{bmatrix} = \mathbf{p}_5$$



Discrete Probability

Markov Chains

A Markov chain: the next state depends only on the current one

Definition

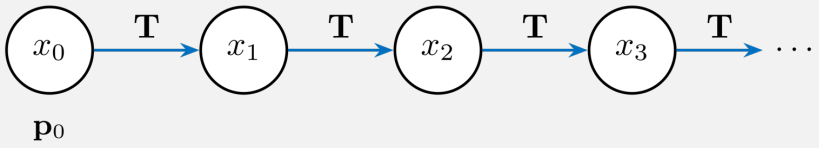
A sequence of random variables x_0, x_1, x_2, … on a state space Ω_x with the Markov property:

$$p(x_k \mid x_{0:k-1}) = p(x_k \mid x_{k-1})$$

It is fully specified by:

initial PMF $(\mathbf{p}_0)_i = p(x_0 = i)$

transition matrix $(\mathbf{T})_{ij} = p(x_k = i \mid x_{k-1} = j)$



Example: the die robot repeating F

Start on face 1 ($p_0 = e_1$); each row is p_k.

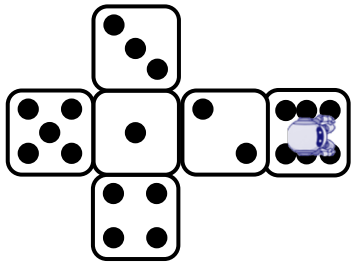


$k = 0$	1	0	0	0	0	0
$k = 1$	.10	.50	.10	.10	.10	.10
$k = 2$	.14	.14	.14	.14	.14	.30
$k = 3$	.16	.16	.16	.16	.22	.16
$k = 4$	.19	.16	.16	.16	.16	.16
$k = 5$	.16	.18	.16	.16	.16	.16
$k = 6$	.17	.17	.17	.17	.17	.17

$k \to \infty$: $\frac{1}{6} \approx .17$ on every face

The peak follows 1→2→6→5→1 →2; slippage spreads it out.

$$\mathbf{T}_F \cdot \mathbf{p}_5 = \begin{bmatrix} 0.1 & 0.1 & 0.1 & 0.1 & 0.5 & 0.1 \\ 0.5 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.5 & 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.5 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.5 \\ 0.1 & 0.5 & 0.1 & 0.1 & 0.1 & 0.1 \end{bmatrix} \cdot \begin{bmatrix} 0.16 \\ 0.18 \\ 0.16 \\ 0.16 \\ 0.16 \\ 0.16 \end{bmatrix} = \begin{bmatrix} 0.17 \\ 0.17 \\ 0.17 \\ 0.17 \\ 0.17 \\ 0.17 \end{bmatrix} = \mathbf{p}_6$$



Hidden Markov Models

A Hidden Markov Model: a Markov chain seen only through noisy sensors

Definition

Hidden states x_0, x_1, … form a Markov chain. We only observe readings, each depending on the current state alone:

$$p(z_k \mid x_{0:k}, z_{1:k-1}) = p(z_k \mid x_k)$$

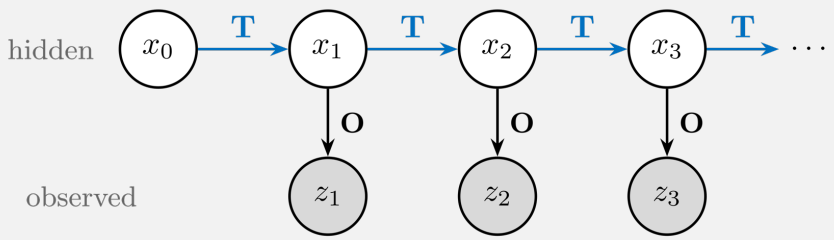
Specified by p_0, the transition matrix T and the observation matrix:

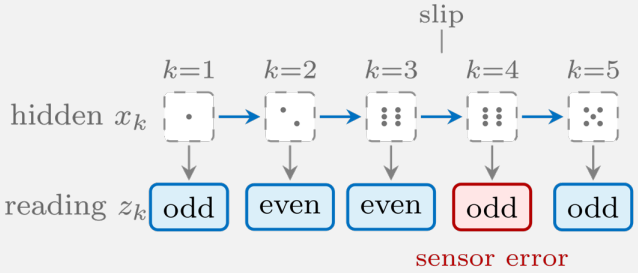
$$(\mathbf{O})_{ij} = p(z_k = j \mid x_k = i)$$

Example: the parity camera

	$z{=}0$ even	$z{=}1$ odd
⚀	0.1	0.9
⚁	0.9	0.1
⚂	0.1	0.9
⚃	0.9	0.1
⚄	0.1	0.9
⚅	0.9	0.1

- Each row of O is a PMF over the two readings.
- Correct parity with probability 0.9, wrong with 0.1.
- The reading never tells the face, only a hint about it.





$$p(x_{0:k}, z_{1:k}) = p(x_0) \prod_{t=1}^{k} p(x_t \mid x_{t-1})\, p(z_t \mid x_t)$$

Filtering: infer the hidden state from the readings so far.

The Recursive Bayes Filter

The belief can be updated without storing the history

Goal: $\mathrm{bel}(x_k) \triangleq p(x_k \mid z_{1:k}, u_{1:k})$

Theorem

$$\overline{\mathrm{bel}}(x_k) = \sum_{x_{k-1}} p(x_k \mid x_{k-1}, u_k)\ \mathrm{bel}(x_{k-1}) \qquad \textbf{prediction}$$

$$\mathrm{bel}(x_k) = \eta_k\, p(z_k \mid x_k)\, \overline{\mathrm{bel}}(x_k) \qquad \textbf{correction}$$

Proof idea

- Correction: Bayes' rule + observation assumption
- Prediction: total probability + Markov assumption

Vector form

$$\overline{\mathbf{p}}_k = \mathbf{T}_{u_k} \mathbf{p}_{k-1},$$

$$\mathbf{p}_k = \eta_k \left(\mathbf{l}_{z_k} \odot \overline{\mathbf{p}}_k\right)$$

Bayes Filter Algorithm

Four lines per time step, at constant cost

Filter(bel(x_{k-1}), u_k, z_k)

1. predict $\overline{\text{bel}}(x_k) = \sum_{x_{k-1}} p(x_k \mid x_{k-1}, u_k)\ \text{bel}(x_{k-1})$
2. weight $\widetilde{\text{bel}}(x_k) = p(z_k \mid x_k)\ \overline{\text{bel}}(x_k)$
3. normalizer $\eta_k = 1 / \sum_{x_k} \widetilde{\text{bel}}(x_k)$
4. normalize $\text{bel}(x_k) = \eta_k\ \widetilde{\text{bel}}(x_k)$

return bel(x_k)

$O(n^2)$

per step: prediction $O(n^2)$, correction $O(n)$.

Constant, however long the robot runs.

Effect on uncertainty

- Prediction spreads the belief.
- Correction concentrates it, on average.

Bayes Filter

Bayes Filter *is the principal* ***algorithm for calculating the belief*** *in robotics.*

1 **Class** *BF*
2 **Method** `Prediction`$(bel(\mathbf{x}_{k-1}), \mathbf{u}_k)$
3 **for** *all* $\mathbf{x}_k \in \mathcal{X}$ **do**
4 $\overline{\text{bel}}(\mathbf{x}_k) = \sum_{\mathbf{x}_{k-1} \in \mathcal{X}} p(\mathbf{x}_k \mid \mathbf{u}_k, \mathbf{x}_{k-1}) \text{bel}(\mathbf{x}_{k-1})$;
5 **return** $\overline{\text{bel}}(\mathbf{x}_k)$;
6 **Method** `Update`$(\overline{bel}(\mathbf{x}_k), \mathbf{z}_k)$
7 **for** *all* $\mathbf{x}_k \in \mathcal{X}$ **do**
8 $\text{bel}(\mathbf{x}_k) = \eta p(\mathbf{z}_k \mid \mathbf{x}_k) \overline{\text{bel}}(\mathbf{x}_k)$;
9 **return** $\text{bel}(\mathbf{x}_k)$;
10 **Method** `Filter`$(bel(\mathbf{x}_{k-1}), \mathbf{u}_k, \mathbf{z}_k)$
11 $\overline{\text{bel}}(\mathbf{x}_k) = \text{Prediction}(\text{bel}(\mathbf{x}_{k-1}), \mathbf{u}_k)$;
12 $\text{bel}(\mathbf{x}_k) = \text{Update}(\overline{\text{bel}}(\mathbf{x}_k), \mathbf{z}_k)$;
13 **return** $\text{bel}(\mathbf{x}_k)$;

Bayes Filter: Worked Example (1/3)



Step 1: an even reading favours Faces 2, 4 and 6

u1 = F **z1 = 0 (even)**

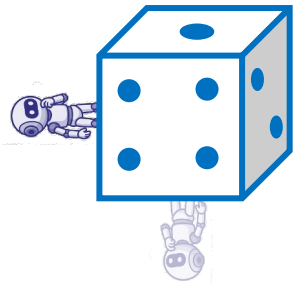
Prediction matrix × vector: $\bar{\mathbf{p}}1 = T_F\ \mathbf{p}0$

$$\underbrace{\begin{bmatrix} .1 & .1 & .1 & .1 & .5 & .1 \\ .5 & .1 & .1 & .1 & .1 & .1 \\ .1 & .1 & .5 & .1 & .1 & .1 \\ .1 & .1 & .1 & .5 & .1 & .1 \\ .1 & .1 & .1 & .1 & .1 & .5 \\ .1 & .5 & .1 & .1 & .1 & .1 \end{bmatrix}}_{\mathbf{T}_F} \underbrace{\begin{bmatrix} .167 \\ .167 \\ .167 \\ .167 \\ .167 \\ .167 \end{bmatrix}}_{\mathbf{p}_0} = \underbrace{\begin{bmatrix} .167 \\ .167 \\ .167 \\ .167 \\ .167 \\ .167 \end{bmatrix}}_{\bar{\mathbf{p}}_1}$$

- Each column of T_F is a PMF.
- Row i collects every way of landing on face i: highlighted, row 2.

row 2: $(\bar{\mathbf{p}}_1)_2 = 0.5 \cdot .167 + 0.1\,(.167 + .167 + .167 + .167 + .167) = .167$

$$u_1 = F\ \ Forward$$



From a uniform belief, moving tells the robot nothing: T_F is doubly stochastic, so every row sums to 1.

Bayes Filter: Worked Example (1/3)



Step 1: an even reading favours Faces 2, 4 and 6

u1 = F **z1 = 0 (even)**

Prediction matrix × vector: $\bar{\mathbf{p}}1 = T_F\ \mathbf{p}0$

$$\underbrace{\begin{bmatrix} .1 & .1 & .1 & .1 & .5 & .1 \\ .5 & .1 & .1 & .1 & .1 & .1 \\ .1 & .1 & .5 & .1 & .1 & .1 \\ .1 & .1 & .1 & .5 & .1 & .1 \\ .1 & .1 & .1 & .1 & .1 & .5 \\ .1 & .5 & .1 & .1 & .1 & .1 \end{bmatrix}}_{\mathbf{T}_F} \underbrace{\begin{bmatrix} .167 \\ .167 \\ .167 \\ .167 \\ .167 \\ .167 \end{bmatrix}}_{\mathbf{p}_0} = \underbrace{\begin{bmatrix} .167 \\ .167 \\ .167 \\ .167 \\ .167 \\ .167 \end{bmatrix}}_{\bar{\mathbf{p}}_1}$$

- Each column of T_F is a PMF.
- Row i collects every way of landing on face i: highlighted, row 2.

row 2: $(\bar{\mathbf{p}}_1)_2 = 0.5 \cdot .167 + 0.1\,(.167 + .167 + .167 + .167 + .167) = .167$

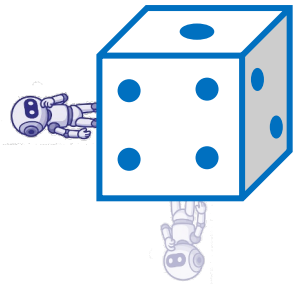
Correction element-wise product ⊙, then normalize: **p1 = η1 (l ⊙ p̄1)**

$$\underbrace{\begin{bmatrix} .1 \\ .9 \\ .1 \\ .9 \\ .1 \\ .9 \end{bmatrix}}_{\mathbf{l}_{z_1=0}} \odot \underbrace{\begin{bmatrix} .167 \\ .167 \\ .167 \\ .167 \\ .167 \\ .167 \end{bmatrix}}_{\bar{\mathbf{p}}_1} = \underbrace{\begin{bmatrix} .017 \\ .150 \\ .017 \\ .150 \\ .017 \\ .150 \end{bmatrix}}_{\mathbf{l} \odot \bar{\mathbf{p}}_1}$$

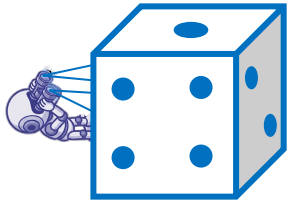
$$\eta_1 = \frac{1}{\sum_i (\mathbf{l} \odot \bar{\mathbf{p}}_1)_i} = \frac{1}{.500} = 2.00$$

$$\mathbf{p}_1 = \eta_1\,(\mathbf{l} \odot \bar{\mathbf{p}}_1) = \begin{bmatrix} .033 \\ .300 \\ .033 \\ .300 \\ .033 \\ .300 \end{bmatrix}$$

$u_1 = F$ *Forward*



$z_1 = 0$ *Even*



From a uniform belief, moving tells the robot nothing: T_F is doubly stochastic, so every row sums to 1.
The even reading then concentrates the belief on the even faces.

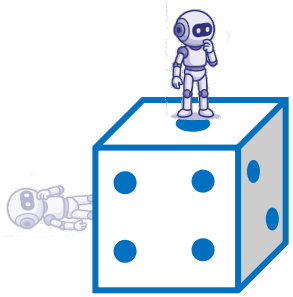
Bayes Filter: Worked Example (2/3)



Step 2: an odd reading singles out Face 5

u2 = F **z2 = 1 (odd)**

$u_2 = F \ Forward$



Prediction matrix × vector: **p̄2 = T_F p1**

$$\underbrace{\begin{bmatrix} .1 & .1 & .1 & .1 & .5 & .1 \\ .5 & .1 & .1 & .1 & .1 & .1 \\ .1 & .1 & .5 & .1 & .1 & .1 \\ .1 & .1 & .1 & .5 & .1 & .1 \\ .1 & .1 & .1 & .1 & .1 & .5 \\ .1 & .5 & .1 & .1 & .1 & .1 \end{bmatrix}}_{\mathbf{T}_F} \underbrace{\begin{bmatrix} .033 \\ .300 \\ .033 \\ .300 \\ .033 \\ .300 \end{bmatrix}}_{\mathbf{p}_1} = \underbrace{\begin{bmatrix} .113 \\ .113 \\ .113 \\ .220 \\ .220 \\ .220 \end{bmatrix}}_{\bar{\mathbf{p}}_2}$$

- Each column of T_F is a PMF.
- Row i collects every way of landing on face i: highlighted, row 5.

row 5: $(\bar{\mathbf{p}}_2)_5 = 0.5 \cdot .300 + 0.1\,(.033 + .300 + .033 + .300 + .033) = .220$

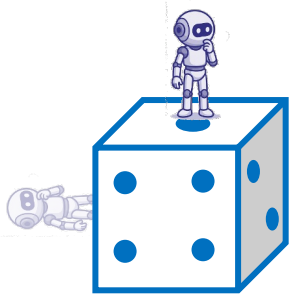
Discrete Probability

Bayes Filter: Worked Example (2/3)

Step 2: an odd reading singles out Face 5

u2 = F **z2 = 1 (odd)**

$u_2 = F \; Forward$



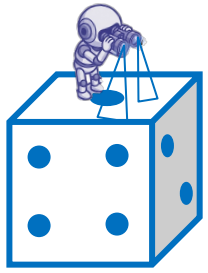
Prediction matrix × vector: $\bar{p}2 = T_F \; p1$

$$\underbrace{\begin{bmatrix} .1 & .1 & .1 & .1 & .5 & .1 \\ .5 & .1 & .1 & .1 & .1 & .1 \\ .1 & .1 & .5 & .1 & .1 & .1 \\ .1 & .1 & .1 & .5 & .1 & .1 \\ .1 & .1 & .1 & .1 & .1 & .5 \\ .1 & .5 & .1 & .1 & .1 & .1 \end{bmatrix}}_{T_F} \underbrace{\begin{bmatrix} .033 \\ .300 \\ .033 \\ .300 \\ .033 \\ .300 \end{bmatrix}}_{p_1} = \underbrace{\begin{bmatrix} .113 \\ .113 \\ .113 \\ .220 \\ .220 \\ .220 \end{bmatrix}}_{\bar{p}_2}$$

- Each column of T_F is a PMF.
- Row i collects every way of landing on face i: highlighted, row 5.

$$\text{row 5: } (\bar{p}_2)_5 = 0.5 \cdot .300 + 0.1\,(.033 + .300 + .033 + .300 + .033) = .220$$

$z_2 = 1 \; Odd$



Correction element-wise product ⊙, then normalize: $p2 = \eta 2 \; (l \odot \bar{p}2)$

$$\underbrace{\begin{bmatrix} .9 \\ .1 \\ .9 \\ .1 \\ .9 \\ .1 \end{bmatrix}}_{l_{z_2=1}} \odot \underbrace{\begin{bmatrix} .113 \\ .113 \\ .113 \\ .220 \\ .220 \\ .220 \end{bmatrix}}_{\bar{p}_2} = \underbrace{\begin{bmatrix} .102 \\ .011 \\ .102 \\ .022 \\ .198 \\ .022 \end{bmatrix}}_{l \odot \bar{p}_2}$$

$$\eta_2 = \frac{1}{\sum_i (l \odot \bar{p}_2)_i} = \frac{1}{.457} = 2.19$$

$$p_2 = \eta_2 \, (l \odot \bar{p}_2) = \begin{bmatrix} .223 \\ .025 \\ .223 \\ .048 \\ .433 \\ .048 \end{bmatrix}$$

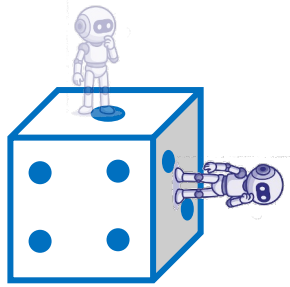
Discrete Probability

Bayes Filter: Worked Example (3/3)

Step 3: prediction and sensor disagree

u3 = F **z3 = 0 (even)**

$u_3 = F \ Forward$



Prediction matrix × vector: $\bar{\mathbf{p}}3 = T_F \, \mathbf{p}2$

$$\underbrace{\begin{bmatrix} .1 & .1 & .1 & .1 & .5 & .1 \\ .5 & .1 & .1 & .1 & .1 & .1 \\ .1 & .1 & .5 & .1 & .1 & .1 \\ .1 & .1 & .1 & .5 & .1 & .1 \\ .1 & .1 & .1 & .1 & .1 & .5 \\ .1 & .5 & .1 & .1 & .1 & .1 \end{bmatrix}}_{\mathbf{T}_F} \underbrace{\begin{bmatrix} .223 \\ .025 \\ .223 \\ .048 \\ .433 \\ .048 \end{bmatrix}}_{\mathbf{p}_2} = \underbrace{\begin{bmatrix} .273 \\ .189 \\ .189 \\ .119 \\ .119 \\ .110 \end{bmatrix}}_{\bar{\mathbf{p}}_3}$$

- Each column of T_F is a PMF.
- Row i collects every way of landing on face i: highlighted, row 1.

row 1: $(\bar{\mathbf{p}}_3)_1 = 0.5 \cdot .433 + 0.1\,(.223 + .025 + .223 + .048 + .048) = .273$

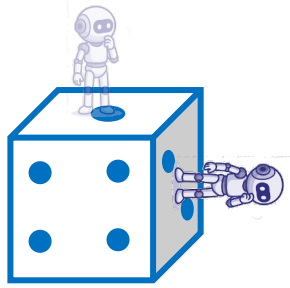
Motion favours the odd Face 1 (5→1);

Bayes Filter: Worked Example (3/3)

Step 3: prediction and sensor disagree

u3 = F **z3 = 0 (even)**

$$u_3 = F \ \ Forward$$



$$z_3 = 0 \ \ Even$$

Prediction matrix × vector: **p̄3 = T_F p2**

$$\underbrace{\begin{bmatrix} .1 & .1 & .1 & .1 & .5 & .1 \\ .5 & .1 & .1 & .1 & .1 & .1 \\ .1 & .1 & .5 & .1 & .1 & .1 \\ .1 & .1 & .1 & .5 & .1 & .1 \\ .1 & .1 & .1 & .1 & .1 & .5 \\ .1 & .5 & .1 & .1 & .1 & .1 \end{bmatrix}}_{\mathbf{T}_F} \underbrace{\begin{bmatrix} .223 \\ .025 \\ .223 \\ .048 \\ .433 \\ .048 \end{bmatrix}}_{\mathbf{p}_2} = \underbrace{\begin{bmatrix} .273 \\ .189 \\ .189 \\ .119 \\ .119 \\ .110 \end{bmatrix}}_{\bar{\mathbf{p}}_3}$$

- Each column of T_F is a PMF.
- Row i collects every way of landing on face i: highlighted, row 1.

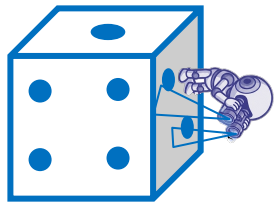
$$\text{row 1: } (\bar{\mathbf{p}}_3)_1 = 0.5 \cdot .433 + 0.1\,(.223 + .025 + .223 + .048 + .048) = .273$$

Correction element-wise product ⊙, then normalize: **p3 = η3 (l ⊙ p̄3)**

$$\underbrace{\begin{bmatrix} .1 \\ .9 \\ .1 \\ .9 \\ .1 \\ .9 \end{bmatrix}}_{\mathbf{l}_{z_3=0}} \odot \underbrace{\begin{bmatrix} .273 \\ .189 \\ .189 \\ .119 \\ .119 \\ .110 \end{bmatrix}}_{\bar{\mathbf{p}}_3} = \underbrace{\begin{bmatrix} .027 \\ .170 \\ .019 \\ .107 \\ .012 \\ .099 \end{bmatrix}}_{\mathbf{l} \odot \bar{\mathbf{p}}_3}$$

$$\eta_3 = \frac{1}{\sum_i (\mathbf{l} \odot \bar{\mathbf{p}}_3)_i} = \frac{1}{.435} = 2.30$$

$$\mathbf{p}_3 = \eta_3\,(\mathbf{l} \odot \bar{\mathbf{p}}_3) = \begin{bmatrix} .063 \\ .392 \\ .044 \\ .247 \\ .027 \\ .228 \end{bmatrix}$$

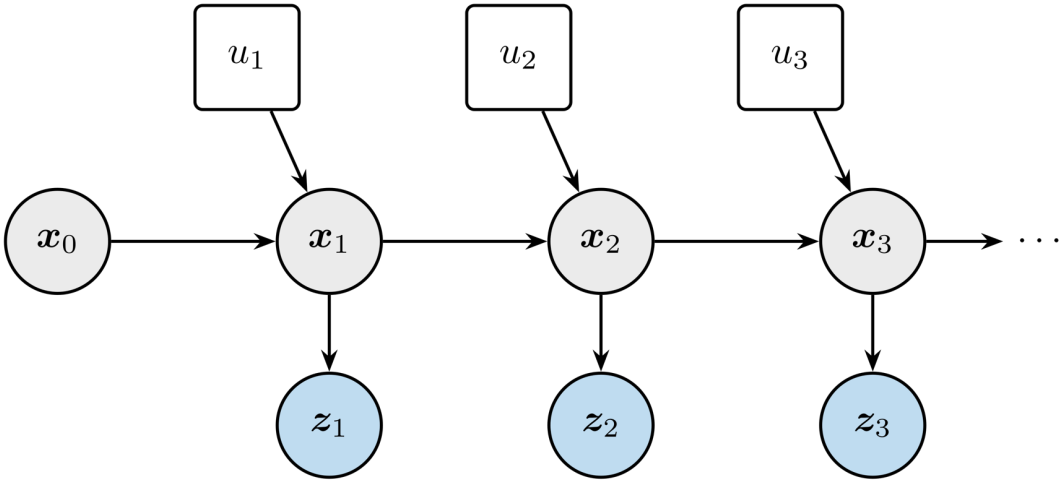


Motion favours the odd Face 1 (5→1); the sensor says even.

The filter arbitrates: Face 2, reached from Face 1 by F, becomes most likely (0.39).

Dynamic Bayesian Network

The DBN is the two HMM assumptions, drawn



How to read it

- Each state x_k has parents x_{k-1} and u_k.
- Each reading z_k has one parent: x_k.
- Arrows point from causes to effects.

Proposition

$$p(x_{0:K}, z_{1:K} \mid u_{1:K}) = p(x_0) \prod_{k=1}^{K} p(x_k \mid x_{k-1}, u_k)\, p(z_k \mid x_k)$$

Summary and Next Steps

From a robot on a die to a general filtering recipe

1	**Language**	Probability spaces, random variables and PMFs describe uncertainty.
2	**Tools**	Product rule, Law of Total Probability and Bayes' rule do most of the work.
3	**Time**	The Markov property, via state augmentation if needed, makes time tractable.
4	**Algorithm**	Localization is an HMM; the Bayes filter alternates prediction and correction.

NEXT:

CONTINUOUS PROBABILITY

An AUV localizing over a seafloor map

- Positions are real-valued
- PMFs become densities, sums become integrals
- Same logic, but no exact filter: histogram, particle and Gaussian filters